

SMAART Minds - Bug Fix Document

Version: 1.0
Date: January 2026
Platform: SMAART Minds - Student Dashboard & Learning Management System

1. Overview

This document catalogs known issues, bugs, and their resolutions for the SMAART Minds platform. It serves as a reference for development and maintenance teams.

2. Bug Classification

Priority Levels

Priority	Response Time	Description
P1 - Critical	Immediate	System down, data loss, security breach
P2 - High	Same day	Major feature broken, significant user impact
P3 - Medium	3 days	Feature degraded, workaround exists
P4 - Low	1 week	Minor issue, cosmetic problems

Status Types

Status	Description
Open	Bug identified, not yet addressed
In Progress	Fix being developed
Resolved	Fix deployed
Closed	Fix verified in production

Won't Fix	Intentional or not a bug
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3. Resolved Issues

3.1 Authentication & Login

BUG-001: First-Time Login Password Change Not Working

Priority: P1

Status: Resolved

Date Fixed: January 2026

Symptoms:

- 400 "Bad Request" error from `/api/auth/first-login-change-password`
- `tempToken` not being properly validated
- Users unable to set new password

Root Cause:

- Missing token validation in password change endpoint
- Incorrect token storage on frontend

Solution:

```
// Backend - routes/auth.js
// Added proper tempToken validation
router.post('/first-login-change-password', async (req, res) => {
  const { tempToken, newPassword } = req.body;

  if (!tempToken) {
    return res.status(400).json({ error: 'Token required' });
  }

  try {
    const decoded = jwt.verify(tempToken, process.env.JWT_SECRET);
    // Continue with password change...
  } catch (err) {
    return res.status(400).json({ error: 'Invalid token' });
  }
});
```

BUG-002: OTP Verification Failure

Priority: P2

Status: Resolved

Date Fixed: December 2025

Symptoms:

- OTP verification returning "Invalid OTP"
- Valid OTPs being rejected

Root Cause:

- OTP expiration time too short
- Race condition in OTP storage

Solution:

- Extended OTP validity to 10 minutes
 - Added atomic OTP verification query
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BUG-003: Session Storage JSON Parsing Error

Priority: P2

Status: Resolved

Date Fixed: January 2026

Symptoms:

- Console error: "Cannot parse undefined"
- Frontend crash on page reload

Root Cause:

- Attempting to parse undefined values from sessionStorage

Solution:

```
// Frontend - Safe JSON parsing
const parseStoredData = (key) => {
  const data = sessionStorage.getItem(key);
```

```
if (!data || data === 'undefined') return null;
try {
  return JSON.parse(data);
} catch (e) {
  return null;
}
};
```

3.2 Registration System

BUG-004: Registration Form Data Not Saving

Priority: P1

Status: Resolved

Date Fixed: January 2026

Symptoms:

- "Failed to save registration" error
- Data lost after submission
- Files not uploading correctly

Root Cause:

- FormData handling issues
- JSON.stringify on file objects
- Backend expecting different field structure

Solution:

```
// Frontend - Proper FormData handling
const formData = new FormData();

// Append files correctly
if (profilePhoto) {
  formData.append('profilePhoto', profilePhoto);
}

// Append JSON data as string
formData.append('registrationData', JSON.stringify({
  personalDetails,
  academicDetails,
  // ...
})));
```

```
// Backend - Proper multipart handling
app.post('/api/users/register-details',
  upload.single('profilePhoto'),
  async (req, res) => {
    const data = JSON.parse(req.body.registrationData);
    // Process data...
  }
);
```

BUG-005: Password Not Being Hashed

Priority: P1

Status: Resolved

Date Fixed: December 2025

Symptoms:

- Passwords stored in plain text
- Security vulnerability

Root Cause:

- Missing bcrypt pre-save hook
- Direct password assignment bypassing hook

Solution:

```
// models/User.js
userSchema.pre('save', async function(next) {
  if (!this.isModified('password')) return next();

  const salt = await bcrypt.genSalt(12);
  this.password = await bcrypt.hash(this.password, salt);
  next();
});
```

3.3 Assessment System

BUG-006: Assessment Progress Not Persisting

Priority: P2

Status: Resolved

Date Fixed: December 2025

Symptoms:

- Answers lost on page refresh
- Users forced to restart assessments

Root Cause:

- Per-question save not implemented
- State only stored in memory

Solution:

- Implemented auto-save after each answer
 - Added resume capability from saved responses
 - Store question order for consistency
-

BUG-007: Question Order Changing on Resume

Priority: P3

Status: Resolved

Date Fixed: December 2025

Symptoms:

- Questions appearing in different order when resuming
- Confusion about which questions were answered

Root Cause:

- Re-shuffling questions on each session load

Solution:

```
// Backend - Preserve question order
if (existingResult && existingResult.questionOrder.length > 0) {
  // Use saved order
  questions = existingResult.questionOrder.map(id =>
    assessmentQuestions.find(q => q._id.equals(id))
  );
}
```

```
} else {  
  // Shuffle for new attempt  
  questions = shuffleArray([...assessmentQuestions]);  
  // Save order  
  result.questionOrder = questions.map(q => q._id);  
  await result.save();  
}
```

3.4 Course System

BUG-008: Module Progress Not Updating

Priority: P2

Status: Resolved

Date Fixed: December 2025

Symptoms:

- Course progress stuck at 0%
- Completed modules not marked

Root Cause:

- Progress calculation error
- Missing update call after module completion

Solution:

```
// Update enrollment progress  
const updateProgress = async (enrollmentId, completedModule) => {  
  const enrollment = await CourseEnrollment.findById(enrollmentId);  
  
  if (!enrollment.modulesCompleted.includes(completedModule)) {  
    enrollment.modulesCompleted.push(completedModule);  
    enrollment.progress = (enrollment.modulesCompleted.length / totalModules) * 100;  
    await enrollment.save();  
  }  
};
```

3.5 UI/UX Issues

BUG-009: 404 Page Styling Broken

Priority: P4

Status: Resolved

Date Fixed: January 2026

Symptoms:

- 404 page not displaying correctly
- Layout issues, elements misaligned

Root Cause:

- CSS class conflicts
- Missing responsive styles

Solution:

- Refactored 404 page CSS
 - Added proper responsive breakpoints
 - Fixed z-index issues
-

BUG-010: Theme Inconsistencies

Priority: P3

Status: Resolved

Date Fixed: January 2026

Symptoms:

- Dark elements appearing on light theme pages
- Text invisible on some backgrounds
- Inconsistent color application

Root Cause:

- Mixed theme references
- Hardcoded colors instead of CSS variables

Solution:

- Converted all colors to CSS variables

- Applied consistent theme classes
 - Added theme context for dynamic switching
-

BUG-011: Vision Board Modal Overflow

Priority: P3
Status: Resolved
Date Fixed: January 2026

Symptoms:

- Modal content extending beyond viewport
- Elements hidden/inaccessible

Root Cause:

- Fixed height not accounting for content
- Missing overflow handling

Solution:

```
.vision-board-modal {  
  max-height: 90vh;  
  overflow-y: auto;  
  padding-bottom: 80px; /* Space for fixed footer */  
}
```

BUG-012: Navigation Menu Issues

Priority: P3
Status: Resolved
Date Fixed: December 2025

Symptoms:

- Duplicate navigation bars
- Menu items not aligned
- Mobile sidebar not closing

Root Cause:

- Multiple navbar component instances
- State not shared between components

Solution:

- Unified navigation into single component
 - Added shared sidebar context
 - Implemented proper mobile menu toggle
-

3.6 Community Features

BUG-013: Community Badge Count Not Resetting

Priority: P4

Status: Resolved

Date Fixed: January 2026

Symptoms:

- Badge showing old post count
- Count not updating after visiting community

Root Cause:

- Last visit timestamp not being updated
- Query not filtering correctly

Solution:

```
// Update last visit on community page load
useEffect(() => {
  const updateLastVisit = async () => {
    await updateUserLastCommunityVisit();
    setBadgeCount(0);
  };
  updateLastVisit();
}, []);
```

3.7 Support Tickets

BUG-014: Ticket Attachment Upload Failing

Priority: P2

Status: Resolved

Date Fixed: December 2025

Symptoms:

- "File upload failed" error
- Attachments not appearing on tickets

Root Cause:

- File size limit too restrictive
- Missing error handling for upload failures

Solution:

```
// Increased limit and added error handling
const upload = multer({
  limits: {
    fileSize: 5 * 1024 * 1024, // 5MB
    files: 3
  },
  fileFilter: (req, file, cb) => {
    const allowed = ['image/jpeg', 'image/png', 'application/pdf', 'text/plain'];
    if (allowed.includes(file.mimetype)) {
      cb(null, true);
    } else {
      cb(new Error('Invalid file type'), false);
    }
  }
}).array('attachments', 3);
```

4. Known Issues (Open)

4.1 Performance Issues

BUG-015: Slow Dashboard Load Time

Priority: P3

Status: Open

Reported: January 2026

Symptoms:

- Dashboard takes 3-5 seconds to load
- Multiple API calls on initial render

Workaround:

- Added loading skeleton UI

Planned Fix:

- Implement data aggregation endpoint
 - Use React Query prefetching
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4.2 Mobile Issues

BUG-016: Video Player Controls Hidden on Mobile

Priority: P3

Status: In Progress

Reported: January 2026

Symptoms:

- Video controls not accessible on small screens
- Touch gestures not working properly

Workaround:

- Use browser's native fullscreen controls

Planned Fix:

- Refactor video player for mobile responsiveness
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5. Troubleshooting Guide

5.1 Common Error Messages

Error Message	Cause	Solution
"Network Error"	Backend not running	Start backend server
"401 Unauthorized"	Token expired	Re-login
"500 Internal Error"	Server exception	Check backend logs
"CORS Error"	Origin not allowed	Check CORS config

5.2 Debug Checklist

1. **Frontend not loading:**
 - Check console for errors
 - Verify `npm run dev` is running
 - Check network tab for failed requests
2. **API calls failing:**
 - Verify backend is running on port 5000
 - Check authentication token
 - Inspect request/response in network tab
3. **Database issues:**
 - Confirm MongoDB is running
 - Check connection string in `.env`
 - Verify database name and collection
4. **File upload issues:**
 - Check uploads/ directory permissions
 - Verify file size limits
 - Check multer configuration

6. Post-Mortem Reports

6.1 Authentication Incident - December 2025

Incident: Users unable to login for 2 hours
Impact: All users affected
Duration: 2 hours
Root Cause: MongoDB connection pool exhausted

- Resolution:**
- Increased connection pool size
 - Added connection health checks
 - Implemented connection retry logic

- Prevention:**
- Set up MongoDB monitoring
 - Added automated alerts for connection issues
 - Implemented graceful connection handling

7. Bug Reporting Template

When reporting a new bug, include the following information:

```
### Bug Title: [Brief description]

**Priority:** P1 / P2 / P3 / P4
**Environment:** Development / Staging / Production
**Browser/Device:** [e.g., Chrome 120, Windows 11]

**Steps to Reproduce:**
1. Step 1
2. Step 2
3. Step 3

**Expected Behavior:**
What should happen

**Actual Behavior:**
What actually happens

**Screenshots/Logs:**
[Attach relevant screenshots or console logs]

**Additional Context:**
Any other relevant information
```

8. Version History

Version	Date	Changes
1.0	Jan 2026	Initial bug documentation

Document End

This document should be updated with each bug fix to maintain an accurate record of platform stability improvements.